

Applied Agentic AI & Generative AI Curriculum

Goal: Job-readiness for "Agentic AI", "Gen AI" and "AI Ops" roles.

Total Duration: ~105 Hours

Format: Project-Based Learning (Multiple Mini-Projects + 3 Major Capstones)

Program Overview

This curriculum is designed to move students from "zero coding knowledge" to building autonomous AI agents. It prioritizes **applied skills**—understanding architectures, orchestration, and tool usage—over deep theoretical mathematics.

MODULE 1: Foundations of AI Scripting & Data

Duration: 32 Hours

Theme: *The New Literacy*. Learning the language of AI (Python) and understanding the raw materials (Neural Nets, NLP).

Submodule 1.1: Python Literacy (Topic 01)

Focus: Writing the "glue code" that connects AI components. No software engineering theory; just practical scripting.

- **Core Concepts:**
 - **Environment:** VS Code, Terminal, Virtual Environments.
 - **Syntax:** Variables, Strings (f-strings), and Basic Data Types.
 - **Logic:** Control Flow (**If/Else** for decision making, **Loops** for batch processing).
 - **Data Structures:** Lists and Dictionaries (The backbone of JSON/API payloads).
 - **Functions:** Writing reusable blocks of code.
- **Mini-Project 1 (Python): "The Log Parser"**
 - *Objective:* Write a script that reads a raw, messy server log file, identifies all lines marked "ERROR", extracts the timestamp, and saves them into a clean report file.

Submodule 1.2: AI Theory & Mechanics (Topics 02, 03)

Focus: Demystifying the "Black Box". Understanding how machines read and learn without the heavy math.

- **Topic: Deep Learning (Topic 02)**
 - *Concepts:* Neural Networks visual intuition, Training vs. Inference, Weights and Biases (conceptual).
 - **Mini-Project 2: "Teachable Machine"**
 - *Objective:* Use a no-code tool (Google Teachable Machine) to train a vision model to distinguish between two objects (e.g., Pen vs. Phone), tweak the "epochs," and export the model.
- **Topic: NLP (Natural Language Processing) (Topic 03)**
 - *Concepts:* Tokenization (How AI sees text), Named Entity Recognition (NER), Sentiment Analysis.
 - **Mini-Project 3: "The Entity Tagger"**
 - *Objective:* Use a Python library ([spaCy](#)) to ingest a news article and automatically extract/print a list of all *People* and *Locations* mentioned in the text.

Module 1 Capstone Project

Project Name: "The JSON Data Cleaner"

- **Time Allocation:** 8 Hours
- **Scenario:** You have received a corrupted dataset of user profiles from a hallucinating AI.
- **Task:** Build a Python script that ingests the messy JSON, validates email formats using logic, standardizes phone numbers, removes duplicates, and exports a "Database-Ready" file.

MODULE 2: The Engine (Models & Memory)

Duration: 35 Hours

Theme: *Architecture.* Mastering Large Language Models (LLMs) and giving them access to private knowledge (RAG).

Submodule 2.1: The Model Layer (Topics 04, 09, 10)

Focus: Interacting with the "Brains"—both proprietary (OpenAI) and open-source.

- **Topic: OpenAI & APIs (Topic 10)**
 - *Concepts:* API Keys security, Rate limits, JSON mode, Temperature settings.
 - **Mini-Project 4: "My First Chatbot"**
 - *Objective:* A script that accepts user input, sends it to the GPT-4 API with a specific "Persona" (System Prompt), and prints the response.
- **Topic: LLM Fundamentals (Topic 04)**
 - *Concepts:* Context Windows, Hallucinations, Token Limits, Prompt Engineering (Few-Shot, Chain of Thought).

- **Mini-Project 5: "The Jailbreak Artist"**
 - *Objective:* A challenge to "trick" an AI with strict safety rules into revealing a hidden password, followed by patching the prompt to secure it.
- **Topic: Hugging Face (Topic 09)**
 - *Concepts:* The Open Source ecosystem, The Model Hub, Running small models locally.
 - **Mini-Project 6: "Local Sentiment Analyst"**
 - *Objective:* Download a small, pre-trained model from Hugging Face and run a script locally to classify a list of movie reviews as "Positive" or "Negative" without using the internet.

Submodule 2.2: The Memory Layer (Topics 06, 11)

Focus: Connecting the AI to your own data (PDFs, Wikis, Docs).

- **Topic: Vector Databases (Topic 11)**
 - *Concepts:* Embeddings (Semantic Math), High-dimensional space, Cosine Similarity.
 - **Mini-Project 7: "The Vibe Matcher"**
 - *Objective:* Store 10 different emotional descriptions in a Vector DB (ChromaDB). Input a new abstract feeling (e.g., "Rainy Sunday") and retrieve the mathematically closest match.
- **Topic: RAG (Retrieval Augmented Generation) (Topic 06)**
 - *Concepts:* The Ingest -> Split -> Embed -> Retrieve pipeline. Chunking strategies.
 - **Mini-Project 8: "One-Page Q&A"**
 - *Objective:* Build a basic pipeline that reads a single text file (e.g., a recipe), embeds it, and allows the user to ask questions specifically about that file.

Module 2 Capstone Project

Project Name: "Docu-Chat (The HR Policy Bot)"

- **Time Allocation:** 10 Hours
- **Scenario:** HR is overwhelmed with questions about the new handbook.
- **Task:** Build a full RAG application. Users upload a PDF Employee Handbook; the bot answers questions ("Can I work from Hawaii?") strictly from the document context and cites the specific page number.

MODULE 3: Agentic Workflows & Orchestration

Duration: 38 Hours

Theme: *Agency.* Moving from passive chatbots to autonomous agents that can plan, use tools, and do work.

Submodule 3.1: Orchestration Frameworks (Topics 05, 07)

Focus: The "Manager" layer that directs the AI's actions.

- **Topic: LangChain (Topic 05)**
 - *Concepts:* Chains, Prompts Templates, Output Parsers, Memory Management.
 - **Mini-Project 9: "The Title Generator"**
 - *Objective:* Create a sequential Chain: Step 1 generates 5 YouTube titles from a topic; Step 2 picks the best one; Step 3 writes a video description for it.
- **Topic: LlamaIndex (Topic 07)**
 - *Concepts:* Data connectors, Specialized Indices for structured vs. unstructured data.
 - **Mini-Project 10: "The Directory Scanner"**
 - *Objective:* Point LlamaIndex at a folder containing mixed text files (invoices, letters, notes) and build a query engine that answers questions across the entire directory.

Submodule 3.2: Production & Deployment (Topics 08, 12)

Focus: User Interfaces and Cloud Deployment.

- **Topic: LLM Apps (Topic 08)**
 - *Concepts:* Building front-ends for AI tools using Streamlit or Gradio.
 - **Mini-Project 11: "The Summarizer UI"**
 - *Objective:* Build a web app with a simple UI: A text box for input and a "Summarize" button that triggers the AI and displays the result.
- **Topic: AWS (Cloud Basics) (Topic 12)**
 - *Concepts:* Cloud vs. Local, Serverless functions (Lambda), API Gateways.
 - **Mini-Project 12: "Hello Cloud"**
 - *Objective:* Deploy a simple Python function to AWS Lambda that returns a JSON greeting, making your code accessible via a public URL.



Module 3 Capstone Project

Project Name: "The Autonomous Market Researcher"

- **Time Allocation:** 14 Hours
- **Scenario:** You need to automate the creation of investment memos.
- **Task:** Build an Agentic Workflow using LangChain/LangGraph:
 1. **Input:** User provides a Company Name (e.g., "Tesla").
 2. **Plan:** Agent decomposes the request into search queries.
 3. **Research (Tool):** Agent uses a Search Tool to find recent news.
 4. **Scrape (Tool):** Agent scrapes content from top URLs.
 5. **Analyze:** LLM summarizes "Bullish" vs. "Bearish" sentiment.

6. **Report:** Agent generates a formatted Markdown Investment Memo and saves it to disk.